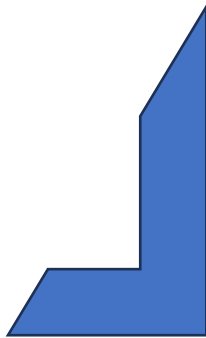


# REDOX TITRATION



Pharmaceutical analysis

By Group one

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To: Inst Feyruza

# Objectives:

**At the end of this presentation, we should be able to:**

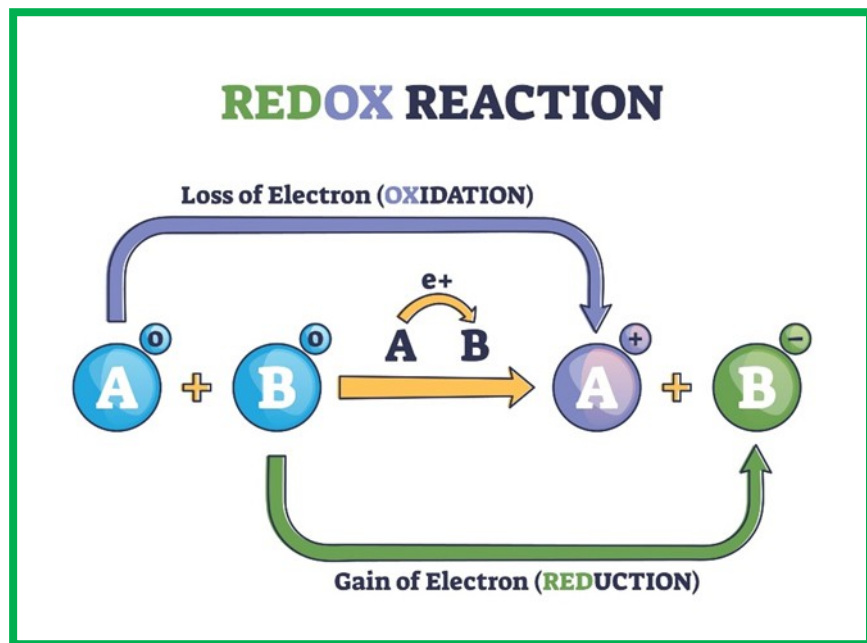
- Define redox titration and explain the fundamental principles involved.
- Define oxidation and reduction.
- Describe the role of indicators in redox titrations and how they function.
- Explain the terms like redox potential.

# What is redox reaction ?

- Oxidation–reduction reactions, commonly known as redox reactions, are reactions that involve the transfer of electrons from one species to another. The species that loses electrons is said to be oxidized, while the species that gains electrons is said to be reduced.

## **Redox= reduction and oxidation**

- **Oxidation:** It can be defined as loss of electrons or increase in oxidation state.
- **Reduction:** It can be defined as gain of electrons or increase of hydrogen content.
- **Oxidizing agent:** substance which get reduced.
- **Reducing agent:** substance which get oxidized



- Further, oxidation and reduction are always found to occur together in a reaction.
- In other words, electron loss of one substance is always accompanied by an electron gain on the part of the other and consequently there are no free electrons in a chemical system.
- Such reactions are termed redox reactions and are also termed **electron transfer reactions**

## Redox titration

**Redox titration** is a method used to determine the concentration of a substance(analyte) in a solution by measuring the volume or concentration of another substance(titrant) that undergoes oxidation-reduction reactions with it.

**A redox reaction can only be used as the basis for a titration if it meets the following requirements:**

1. The redox reaction must be rapid and nearly complete (<99% success is intolerable).
2. The end point must be measurable or detectable, either by means of a color indicator or potentiometry.
3. The process for the exchange of electrons must be stoichiometric, meaning that the pairs of redox systems (oxidizing and reducing agents) are appropriate and equal.

# Principle of Redox Titration

- Redox titration works on the principle that in the oxidation process electrons are lost, while in the reduction process, electrons are gained. Transfer of electrons occurs between titrant and analyte.
- The redox titration is carried out by adding a known amount of oxidizing or reducing agent to a sample of the unknown species and then measuring the oxidizing or reducing agent's concentration change. The unknown species' concentration is calculated using the change in concentration

- **Titration Endpoint:** The endpoint is determined by a slight change in the system, like a color change or appearance of a precipitate, indicating that the reaction is complete.
- **Equivalence Point:** At this point, the amount of titrant added is equal to the amount of analyte present, marking the completion of the reaction.
- **Titration Curves:** These curves show pH or potential changes as titrant volume increases, reflecting the reaction's progress.
- **Choice of Indicator:** Indicators are selected based on their ability to change color near the equivalence point.
- **Stoichiometry:** Balanced chemical equations help calculate the amount of titrant needed for the reaction.

# Redox Potential

- Redox potential (also known as oxidation / reduction potential) is a measure of the tendency of a chemical species to acquire electrons from or lose electrons to an electrode and thereby be reduced or oxidized respectively.
- Redox potential is expressed in **volts (V)**.
- Each species has its own intrinsic redox potential; for example, the more positive the reduction potential (reduction potential is more often used due to general formalism in electrochemistry), the greater the species' affinity for electrons and tendency to be reduced.

# How does redox potential work ?

- An inert electrode, it is typically made of platinum, although gold and graphite can be used as well is placed in the redox system. It acts as a platform for electron transfer.
- If the system tends to act as an oxidizing agent i.e., it will take electrons from platinum, leaving the latter positively charged.
- On the other hand, if the system has reducing properties i.e., electrons will be given to the metal, which will acquire a negative charge.

- The magnitude and sign of the potential on the platinum electrode will thus be a measure of the oxidizing or reducing properties of the system.
- These redox potentials enable us to predict which ion will oxidize or reduce other ions at unit activity (or molar concentration).

# Redox indicator

- Redox indicators are substances that change color depending on the redox potential of the solution. They are crucial in redox titrations to visually detect the endpoint.

## Redox indicators

| Indicators                    | Colour change |              |
|-------------------------------|---------------|--------------|
|                               | Oxidised Form | Reduced form |
| ferroin                       | Pale blue     | red          |
| Diphenyl benzidine            | Violet        | Colourless   |
| Diphenyl amine                | Violet        | Colourless   |
| Nitro ferroin                 | Pale blue     | red          |
| 2,6-dibromo phenol indophenol | Blue          | Colourless   |
| 2,6-dibromo phenol indophenol | Blue          | Colourless   |
| 2,6-dibromo phenol indophenol | Blue          | Colourless   |

- The oxidized and reduced forms of the indicator have different colors.

# Types of the redox indicator

## A. Titrant as indicator(Self-indicator)

When the titrant of intense color is used, the end point can be detected by the color imparted to the solution by the first excess of titrant.

### E.g. $\text{KMnO}_4$

- Potassium permanganate is deep purple. A dilute solution of potassium permanganate is pink. The product of its reduction,  $\text{Mn}^{2+}$ , is extremely faint pink, nearly colorless.

- During a titration with potassium permanganate, the purple color of the  $\text{MnO}_4^-$  is removed as soon as it is added because it is reduced to  $\text{Mn}^{2+}$ .
- As soon as the titration is complete, a fraction of a drop of excess  $\text{MnO}_4^-$  solution imparts a definite pink color to the solution, indicating that the reaction is complete.
- The volume of  $\text{KMnO}_4$  required to reach the endpoint is known as the endpoint volume.

## B. Starch indicator

- This indicator is used for titration involving iodine.
- Starch forms a complex with triiodide ion or iodine ion that is dark blue color. When a starch is added to iodide it give a colorless solution.

### **To better understand this:**

1. In the burette there is the reducing agent, which is the thiosulphate in sodium thiosulphate form.
2. In the flask, we have the iodine solution ,which is the oxidizing agent and this solution contains starch which is the indicator.

3. Initially, the solution in the flask is blue-black because there is only iodine solution in it.
4. During titration, we add the reducing agent to react with iodine solution and since the iodine concentration is still greater than the iodide, the color is still blue-black
5. At the equivalent point, all the iodine solution is consumed and only the iodide remain resulting in a colorless solution.

### C. True- redox indicators (internal indicators)

- Internal indicators are added directly to the titration solution before the titration begins. They participate in the redox reaction and undergo a color change at the endpoint.

#### Examples:

- ❖ **Diphenylamine:** Used in titrations involving potassium dichromate. It changes color from colorless to violet-blue at the endpoint.
- ❖ **Ferroin:** Used in titrations involving ceric sulfate. It changes color from red to pale blue at the endpoint.

## D. External indicator

- External indicators are not added directly to the titration solution but are used to detect the endpoint indirectly. A small drop of the titrated solution is withdrawn and mixed with a drop of the indicator on a white tile or spot plate.
- **Potassium Ferrocyanide:** Used in titrations involving potassium permanganate. It forms a blue complex with ferric ions, indicating the endpoint.

## E. Potentiometric Method

- It is a physicochemical method where the equivalence point is detected by large changes in potential.
- This method can be used when suitable indicators are not available, visual method is of limited accuracy (colored solutions or very dilute solutions).
- Potential is measured by a potentiometer.

# Advantages and disadvantages of redox titration

## **Advantages of redox titration**

1. As little equipment is needed, redox titration is typically inexpensive.
2. It does not necessitate a great level of expertise.
3. It can be easily performed in laboratory.
4. The process is frequently quick.

## **Disadvantages of redox reaction**

1. It is destructive method usually require large quantity of analytes.
2. The reaction is takes place in solution phase which may not be suitable for analysis of some compounds.
3. It has low accuracy.

# Application

Redox titrations are commonly used in various fields, including:

- environmental analysis (e.g., determining the concentration of oxygen in water),
- food industry (e.g., measuring vitamin C), and
- pharmaceuticals (e.g., assessing the purity of compounds).

# References

1. Microsoft Word - Unit\_10\_\_for\_CRCfinal[1].doc
2. <https://scienceinfo.com/redox-indicators/>
3. <https://www.quora.com/What-is-the-definition-of-self-indicator>
4. <https://www.vedantu.com/question-answer/difference-between-external-indicators-and-class-11-chemistry-cbse-60d76f8f4fd8652fb1b8c43b>
5. <https://www.scribd.com/document/461271113/redox-indicators-pdf>
6. [https://uomustansiriyah.edu.iq/media/lectures/4/4\\_2019\\_04\\_11!10\\_48\\_41\\_AM.pdf](https://uomustansiriyah.edu.iq/media/lectures/4/4_2019_04_11!10_48_41_AM.pdf)
7. <https://unacademy.com/content/nta-ugc/study-material/pharmaceutical-analysis/theory-and-principles-of-redox-titration-you-need-to-know/>
8. <https://www.geeksforgeeks.org/redox-titration/>
9. Pharmaceutical analysis by pd chaithanya Sudha z lib org.pdf

THANK  
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